Loan Default Prediction

An interpretable credit-scoring model for home equity loan approvals

Capstone Project | Practical Data Science | MIT Professional Education - Applied AI and Data Science Program

Shubham Garg | November 2026

Key takeaways

1 in 5

home equity loans in the data ended in default. Accuracy alone is a misleading yardstick.

62%

default rate when the applicant's debt-to-income ratio is missing, versus 8.6% when it is recorded. Unverified income is the strongest warning sign.

93%

of future defaulters are caught by the proposed Random Forest model, while 2 of every 3 applicants it flags really would have defaulted.

~$9M

indicative annual benefit on 6,000 applications, from roughly $10.8M of default-related cost down to about $1.8M.

The problem: approving home equity loans without losing the profit to defaults

- Interest on home loans is a major share of retail bank profit; bad loans eat it up.
- Today each application is judged manually: slow, inconsistent across underwriters, open to error and bias.
- The consumer credit department wants an empirically derived credit-scoring model built on its own recent loans.
- Constraint: under the Equal Credit Opportunity Act every rejection needs a specific, defensible reason, so the model must be interpretable.

Objective: predict which applicants are likely to default and tell the bank which attributes to weigh when approving a loan.

The data: Home Equity dataset (HMEQ)

5,960
recent loans

12
attributes known at application

19.9%
defaulted or seriously delinquent

44%
of applications have a blank field

Approach: from raw applications to a decision the bank can defend

1

Explore

Univariate, bivariate and correlation analysis; default rate by band; pattern of missing values

2

Prepare

Flag every missing field, impute median/mode, cap outliers for the linear model only, one-hot encode, 70/30 stratified split

3

Model

Logistic Regression (linear benchmark), Decision Tree, Random Forest; class weights; cross-validated grid search

4

Measure

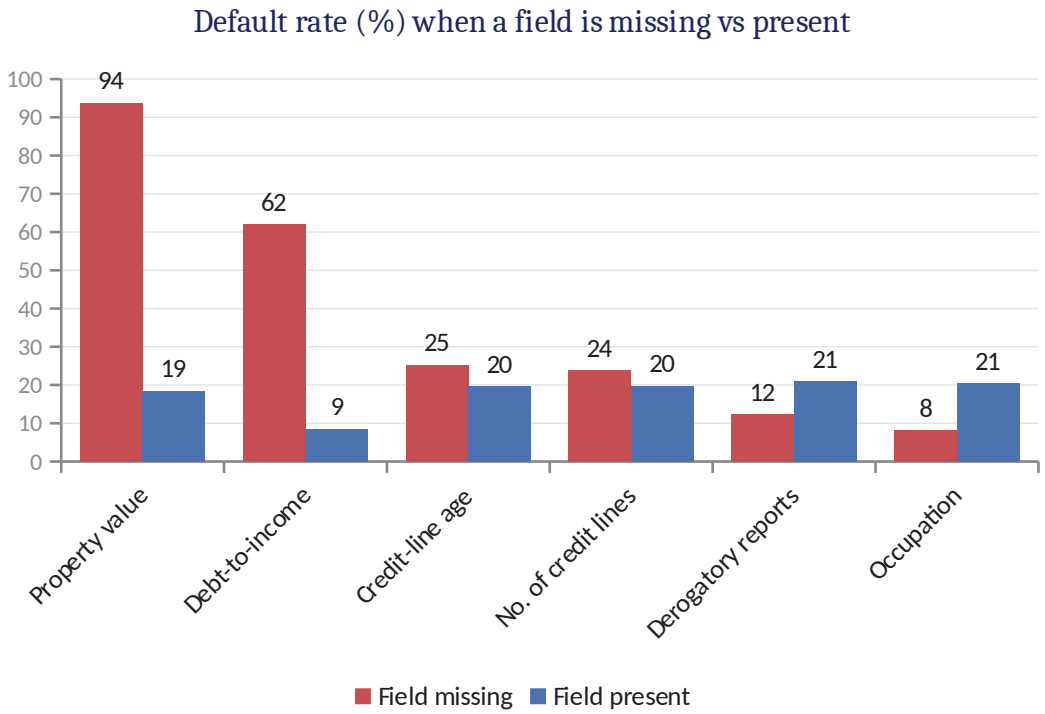
Recall on defaulters as the primary metric; precision, F1, ROC-AUC as guard-rails; cost-based threshold

5

Recommend

Model specification, business rules, implementation plan, benefits, risks

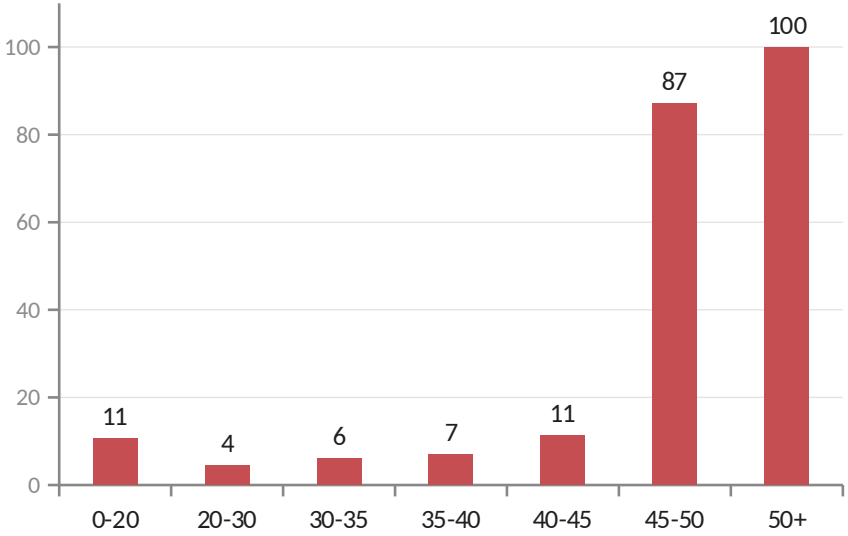
Insight 1: information the applicant did not provide is the strongest warning sign



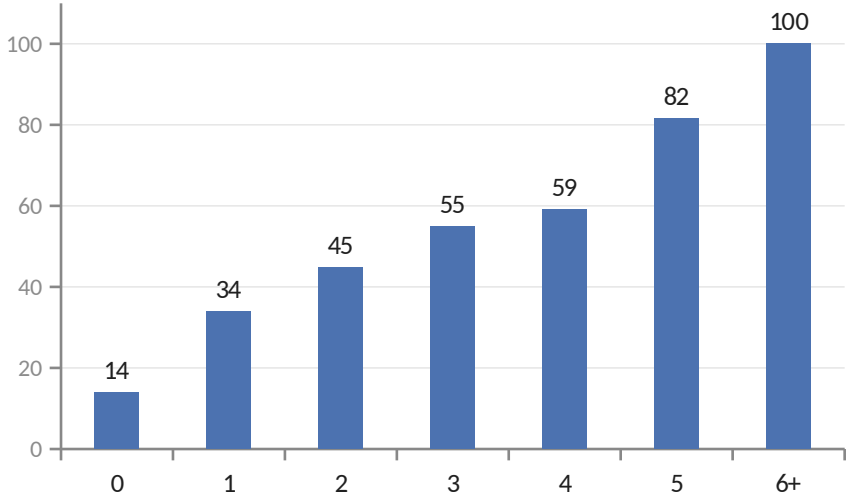
- No recorded debt-to-income ratio: 62% default (1,267 applicants, 21% of the data).
- No property valuation: 94% default (112 applicants).
- Overall default rate is 20%, so these gaps multiply risk three to five times.
- Interpretation: income or collateral that could not be verified, or that the applicant chose not to supply.
- Action: every imputed field gets a 'was missing' flag in the model, and verification should become mandatory in the process.

Insight 2: debt burden has a cliff at 45%, and past delinquency repeats

Default rate (%) by debt-to-income ratio band (%)



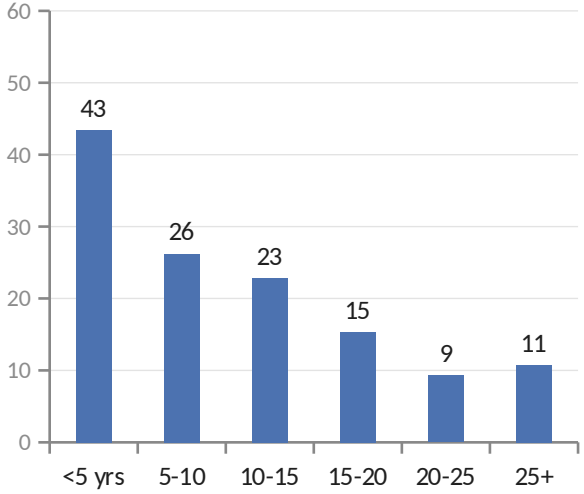
Default rate (%) by number of delinquent credit lines



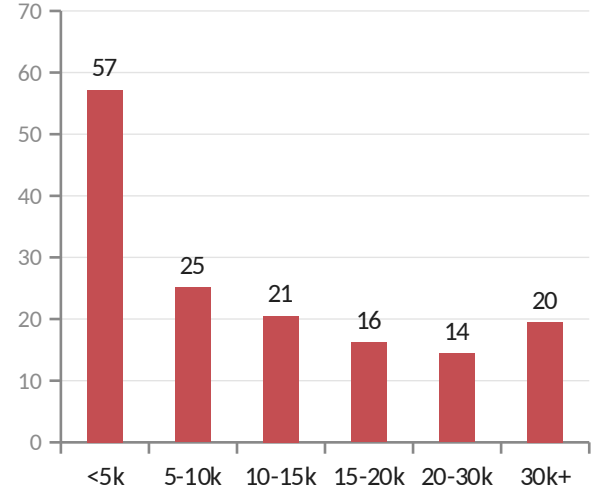
Below a 40% ratio only 4-7% default; above 45% almost everyone does. Each delinquent line or derogatory report roughly doubles the risk (0 reports: 17%, 1: 39%, 2: 51%, 3+: 70%+). Both are plain-language rules an applicant can be given.

Insight 3: long credit histories protect, small loans do not, occupation matters less

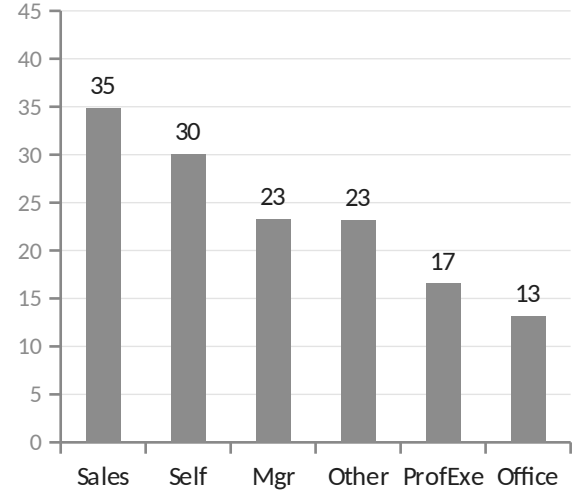
By age of oldest credit line



By loan amount ($)



By occupation



Default rate (%) by group. Applicants with under five years of credit history default four times as often as those with twenty-plus years. Loans under $5,000 default 57% of the time. Occupation shifts risk moderately but adds almost nothing once financial behaviour is known, so the final model does not need it.

Measure of success: the two mistakes do not cost the same

Missed defaulter

The model says 'will repay', the customer defaults.

Bank loses most of the principal plus collection cost: about $9,000 on a typical $18,600 loan.

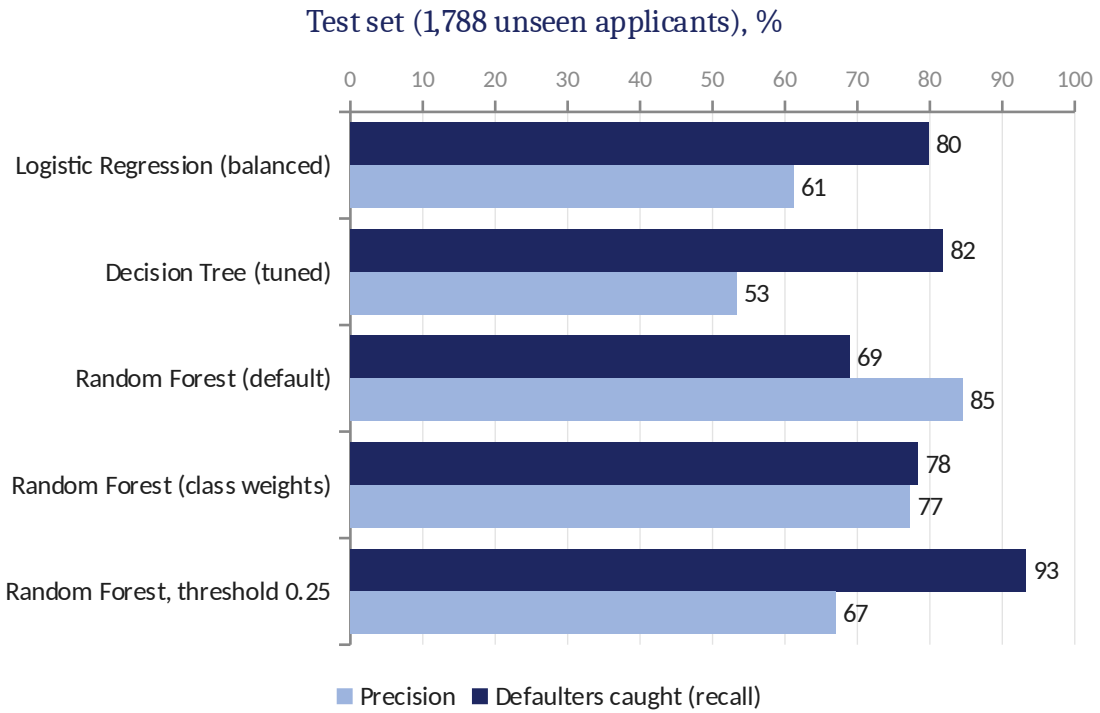
Wrongly declined

The model says 'will default', the customer would have repaid.

Bank forgoes the interest margin on one good customer: about $2,000.

- A missed defaulter costs roughly 4.5 times a wrongly declined customer, so the primary metric is Recall: the share of future defaulters the model catches.
- Guard-rails: Precision (how many of the flagged applicants really default) and F1, so the model does not simply decline everyone. ROC-AUC measures how well applicants are ranked by risk, which matters when the threshold moves.
- All models are tuned by cross-validated grid search on recall and judged on a held-out 30% of applicants the models never saw.

Model comparison: the Random Forest gives the best balance



- Linear vs non-linear: logistic regression reaches 80% recall but only 61% precision. The data is made of thresholds, which trees capture and a straight line cannot.
- Single tree vs forest: the tuned tree catches 82% but declines almost one good customer per defaulter caught (53% precision).
- The class-weighted forest catches 78% at 77% precision, with the best ROC-AUC (0.96): it ranks applicants by risk most accurately.
- Because it ranks so well, lowering its threshold to 0.25 lifts recall to 93% at 67% precision, the lowest expected cost of any option.

Proposed solution: a Random Forest scorer with a Decision Tree explanation layer

Model specification

- Scoring engine: Random Forest, 200 trees, class weights 0.2 (repaid) / 0.8 (default).
- Inputs: the 10 financial attributes plus 'debt-to-income missing' and 'property value missing' flags. Occupation and loan purpose are not needed.
- Decision threshold 0.25 (cost-minimising) instead of the default 0.5; approve / refer / decline bands keep an underwriter in the loop near the threshold.
- Explanation layer: a pruned Decision Tree built on the same data. Its rules (ratio missing, ratio above 44%, delinquencies, derogatory reports, credit-history length) supply the adverse-action reasons the ECOA requires.
- Validated on a 30% hold-out; hyper-parameters chosen by stratified cross-validation.

93%
of defaulters caught

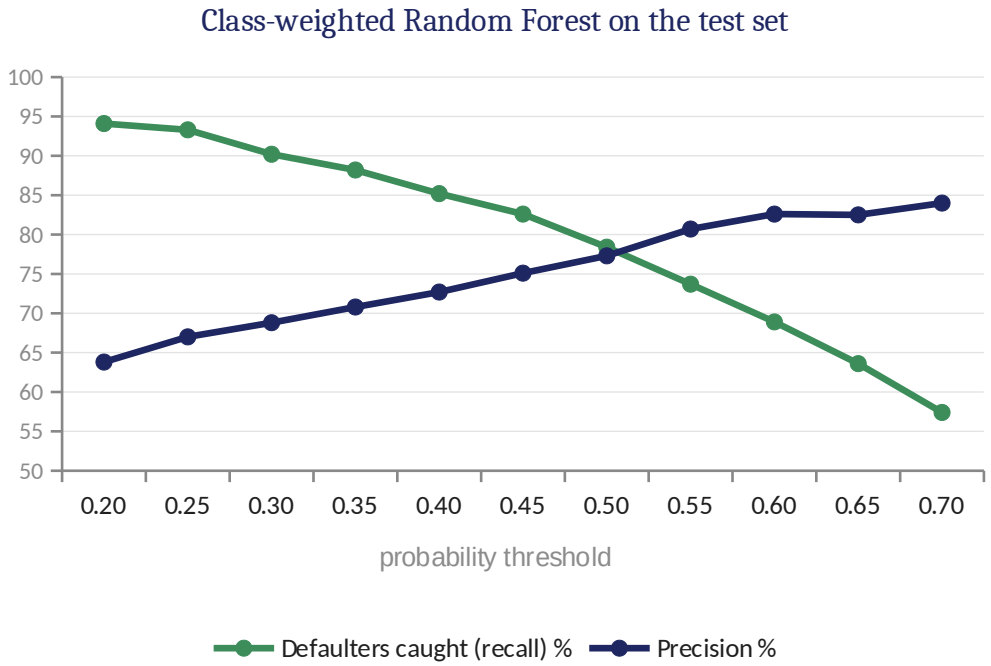
67%
of flagged applicants truly default

0.96
ROC-AUC: risk ranking quality

90%
overall accuracy

Per 1,000 applicants: about 187 of 200 defaulters flagged in advance, 13 missed, 92 good customers referred or declined.

Why a 0.25 threshold: the decision cut-off is a business dial, not a constant



At 0.50 (default)

- 43 missed defaulters
- 46 good customers declined
- $479k indicative cost

At 0.25 (proposed)

- 13 missed defaulters
- 92 good customers declined
- $304k indicative cost

Per 1,000 applicants; cost assumes $9,000 per missed defaulter and $2,000 per wrongly declined customer. Thresholds 0.20-0.30 are all within 12% of the minimum, so the choice is robust.

Business impact: an indicative $9 million a year

Per year, 6,000 applications	Current process	With the proposed model
Loans that default	about 1,200 (20%)	about 80 slip through
Default-related losses	about $10.8M	about $0.7M
Good customers declined or referred	none	about 550, margin forgone about $1.1M
Total cost of credit errors	**about $10.8M**	**about $1.8M**

~$9M

indicative gross annual benefit; about $8.5M after running costs

- Assumptions: 6,000 applications a year, average loan $18,600, loss given default about 50%, forgone margin about $2,000 per declined good customer, production performance equal to test performance.
- Costs: roughly $300-400k one-off (pipeline, integration, validation, fairness review, training) and about $100k a year to run and monitor. Pays back within the first year even if only a quarter of the benefit is realised.
- Non-financial: faster decisions, consistent treatment, auditable reasons for every decline, verified income and valuation on every application.

Recommendations and implementation roadmap

1 Months 0-2: prepare

- Make income and valuation verification mandatory fields
- Build the scoring pipeline and reason codes
- Calibrate the threshold with real loss and margin figures
- Fairness and model-risk review

2 Months 2-4: shadow and pilot

- Score every application alongside the manual decision
- Compare outcomes, tune the approve / refer / decline bands
- Train underwriters on reason codes and referrals

3 Month 4 onwards: scale

- Switch on automated approve / decline for clear cases
- Monthly monitoring of recall, precision, approval rate, missing fields
- Yearly re-training; threshold review with finance

Policy rules to adopt now, independent of the model

- Decline or escalate when the debt-to-income ratio is above 45%, when there are two or more delinquent credit lines, or two or more major derogatory reports.
- Treat an application with no verifiable income or property valuation as high risk, never as neutral.

Risks, challenges and further analysis

Risks and mitigations

- Selection bias: data covers approved loans only. Mitigate with a shadow period on all applications and reject inference.
- The 'ratio missing' signal will change once verification is mandatory. Re-train after the rule is in place.
- Economic drift: monitor monthly, re-train yearly, review the threshold.
- Fair lending: keep occupation out of the model; disparate-impact testing before launch and annually.
- Explainability and automation bias: tree-based reason codes, a referral band, documented overrides.

Further analysis

- Reject inference to cover applicants the old process declined.
- Engineered features: loan-to-value and payment-to-income ratios.
- Gradient boosting and probability calibration, so a score of 0.25 means a 25% chance of default.
- Time-based validation once application dates are available.
- Risk-based pricing for the referral band; early-warning scoring on existing loans.
- Per-applicant SHAP-style reason codes in applicant-friendly language.

Thank you

Questions and discussion

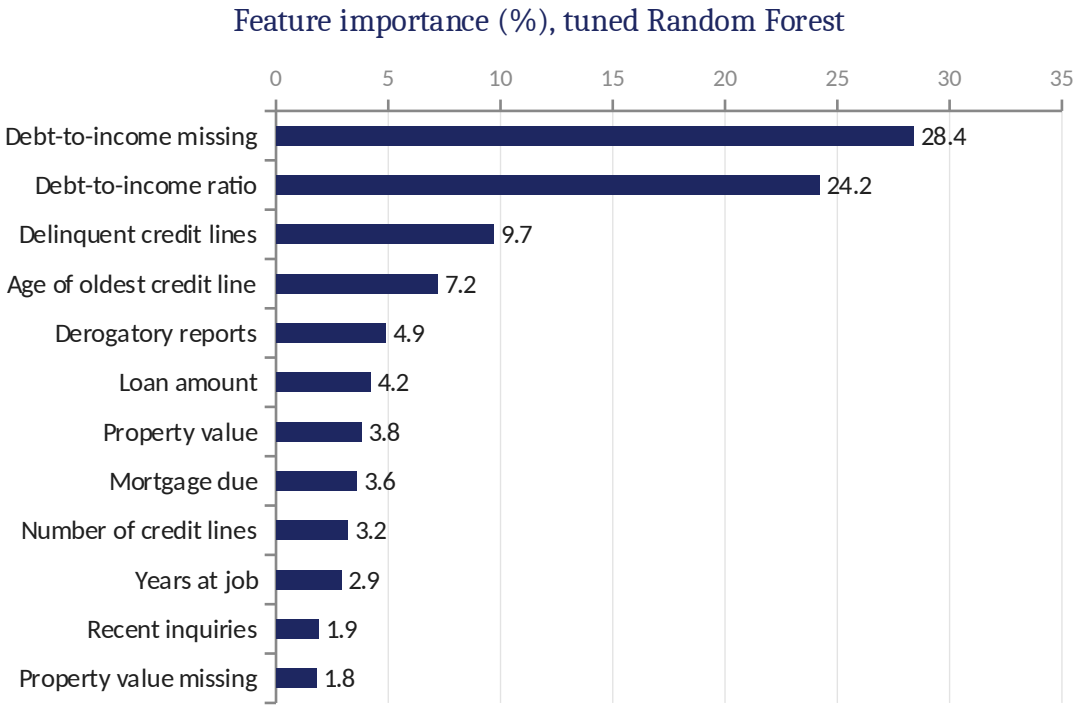
Catch 93% of defaulters, explain every decision, and save an indicative $9M a year.

Appendix A: all models on the held-out test set

Model	Recall	Precision	F1	Accuracy	ROC-AUC	Train recall
Logistic Regression (plain)	0.569	0.772	0.655	0.880	0.907	0.643
Logistic Regression (balanced)	0.798	0.612	0.693	0.859	0.906	0.823
Decision Tree (default)	0.686	0.706	0.696	0.880	0.807	1.000
Decision Tree (class weights)	0.616	0.728	0.668	0.878	0.779	1.000
Decision Tree (tuned: balanced, min 20 per leaf)	0.818	0.533	0.645	0.820	0.879	0.924
Random Forest (default, 200 trees)	0.689	0.845	0.759	0.913	0.962	1.000
Random Forest (class weights 0.2/0.8)	0.784	0.773	0.779	0.911	0.958	1.000
Random Forest (tuned for recall)	0.784	0.653	0.712	0.874	0.933	0.921
Random Forest (tuned, 12 features)	0.782	0.664	0.718	0.878	0.928	0.917
Proposed: RF class weights, threshold 0.25	**0.933**	**0.670**	**0.780**	**0.895**	**0.958**	-

Train recall of 1.000 marks the un-pruned models that memorised the training data. Grid search used stratified cross-validation on the training set only, scoring on recall.

Appendix B: what the Random Forest relies on



- The debt-to-income ratio and its missing flag carry more than half (53%) of the forest's decision power.
- Next come repayment history (delinquencies, derogatory reports) and depth of credit history.
- Loan, property and mortgage amounts matter in combination but are weak on their own.
- Occupation and loan-purpose dummies rank at the bottom; removing them and nine minor flags left performance unchanged (12-feature model).
- Logistic-regression coefficients and the decision-tree rules point to the same drivers, so the signal is stable across techniques.

Appendix C: the explanation layer, rules read from the pruned Decision Tree

1	**Is the debt-to-income ratio missing?**	Yes: highest-risk branch (62% historical default). Verify income or decline.
2	**If recorded, is the ratio above about 44%?**	Yes: almost certain default in the data (87-100%). Decline.
3	**Any delinquent credit lines?**	One or more pushes the applicant back into the risky branch, even with a healthy ratio.
4	**Any major derogatory reports, and how long is the credit history?**	Short history (under about 15 years) with a derogatory report is risky; long clean histories are safe.
5	**Loan size, mortgage balance and property value**	Refine the decision deeper in the tree; small loans with weak history lean towards default.

Tuned tree: Gini criterion, balanced class weights, minimum 20 applicants per leaf, depth 14, 105 leaves. Test recall 82%, precision 53%: used for explanations, not for scoring.

Appendix D: data treatment and pre-processing

Step	What was done	Why
Duplicates	None found; no rows dropped	5,960 loans retained, including 2 with only the loan amount recorded
Missing values	A 0/1 'was missing' flag per column, then median (numeric) or mode (categorical) imputation	Missingness of debt-to-income and property value is the strongest default signal; dropping rows would lose 44% of the data
Outliers	IQR capping for the linear model only; DEROG and DELINQ capped at the 99th percentile; tree models use raw values	Linear models are pulled by extremes (debt-to-income 203%, credit history 97 years); trees split on rank order and keep the 'above 50% always defaults' signal
Encoding and scaling	One-hot encoding of REASON and JOB (drop first); standard scaling for the linear model only	Comparable coefficients and stable convergence for logistic regression
Split	70% train / 30% test, stratified on the target, fixed seed; identical applicants in both data versions	Both sets keep the 20% default rate; results comparable across models
Imbalance	Class weights (balanced or 0.2/0.8) inside the models rather than resampling	Keeps every real applicant; lifted recall by 23 points for logistic regression and 10 for the forest